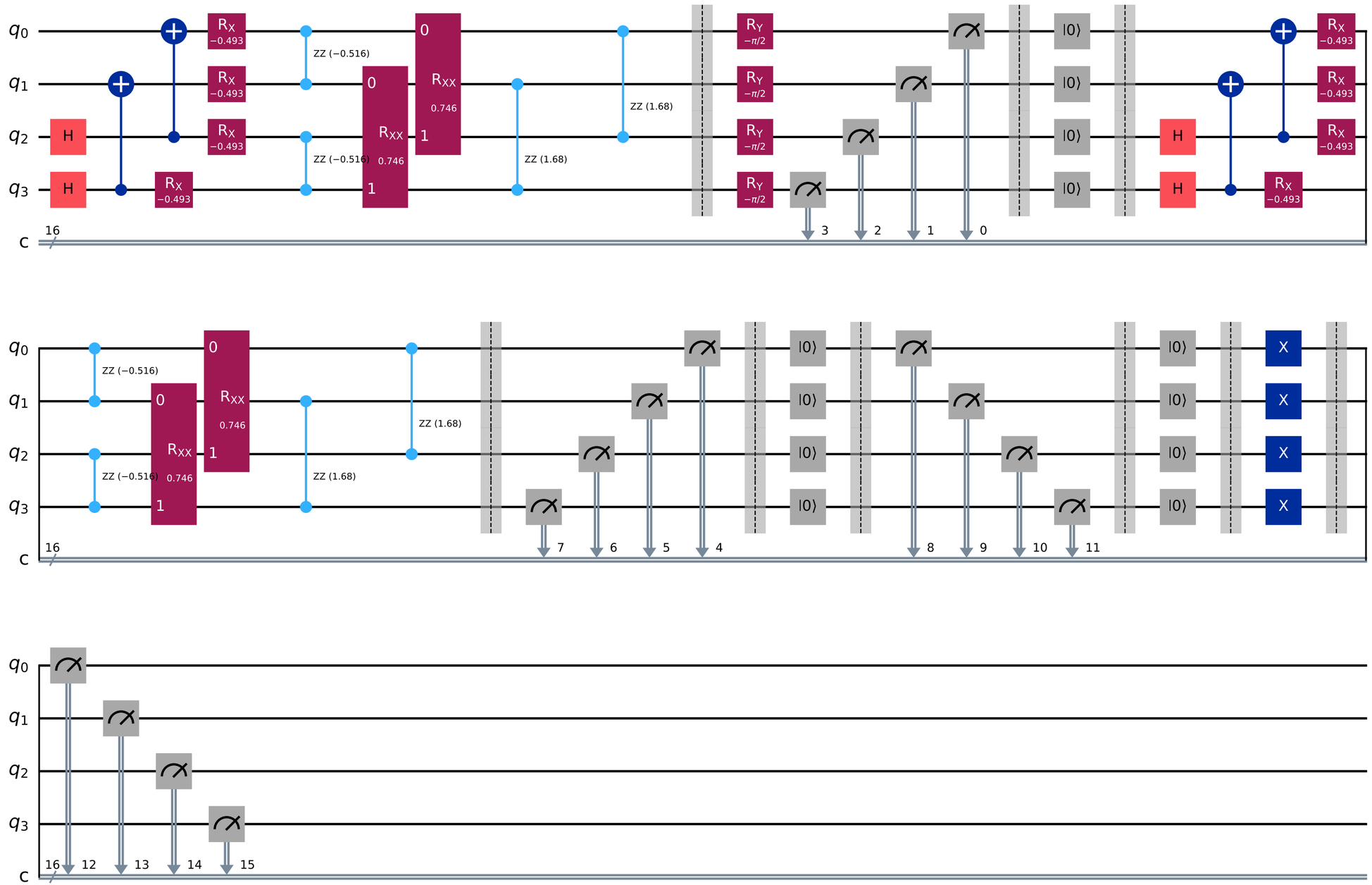
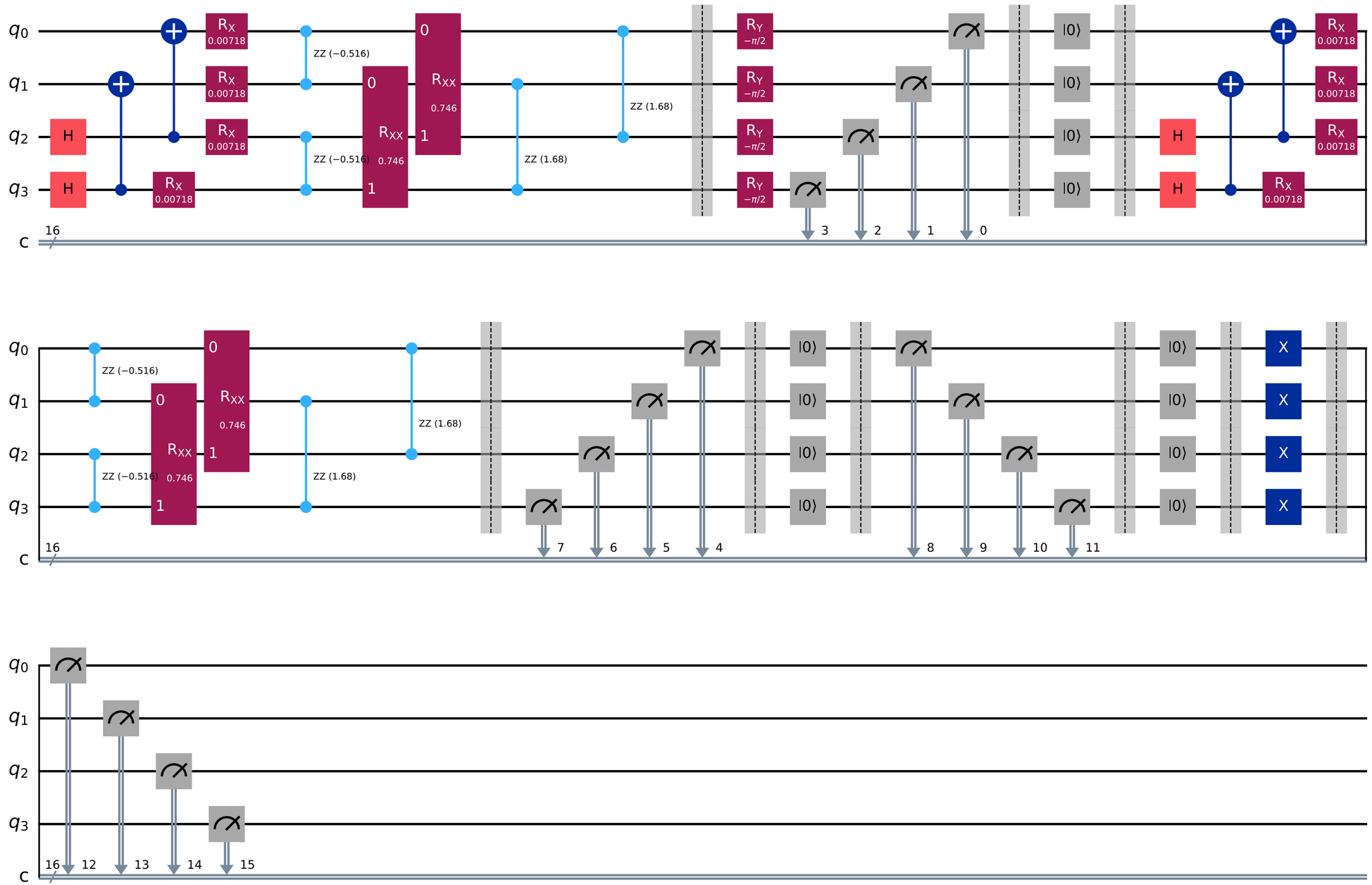
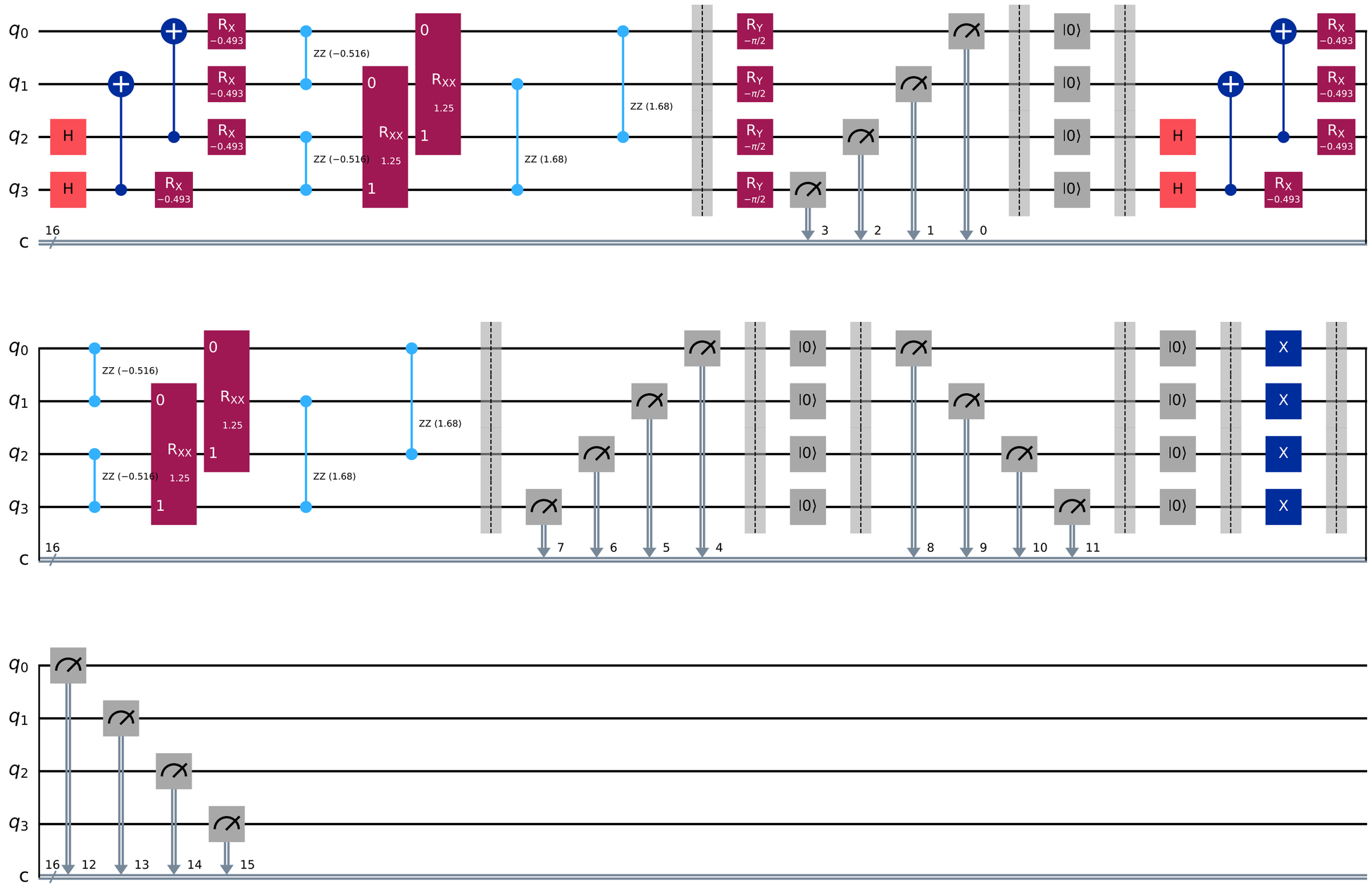
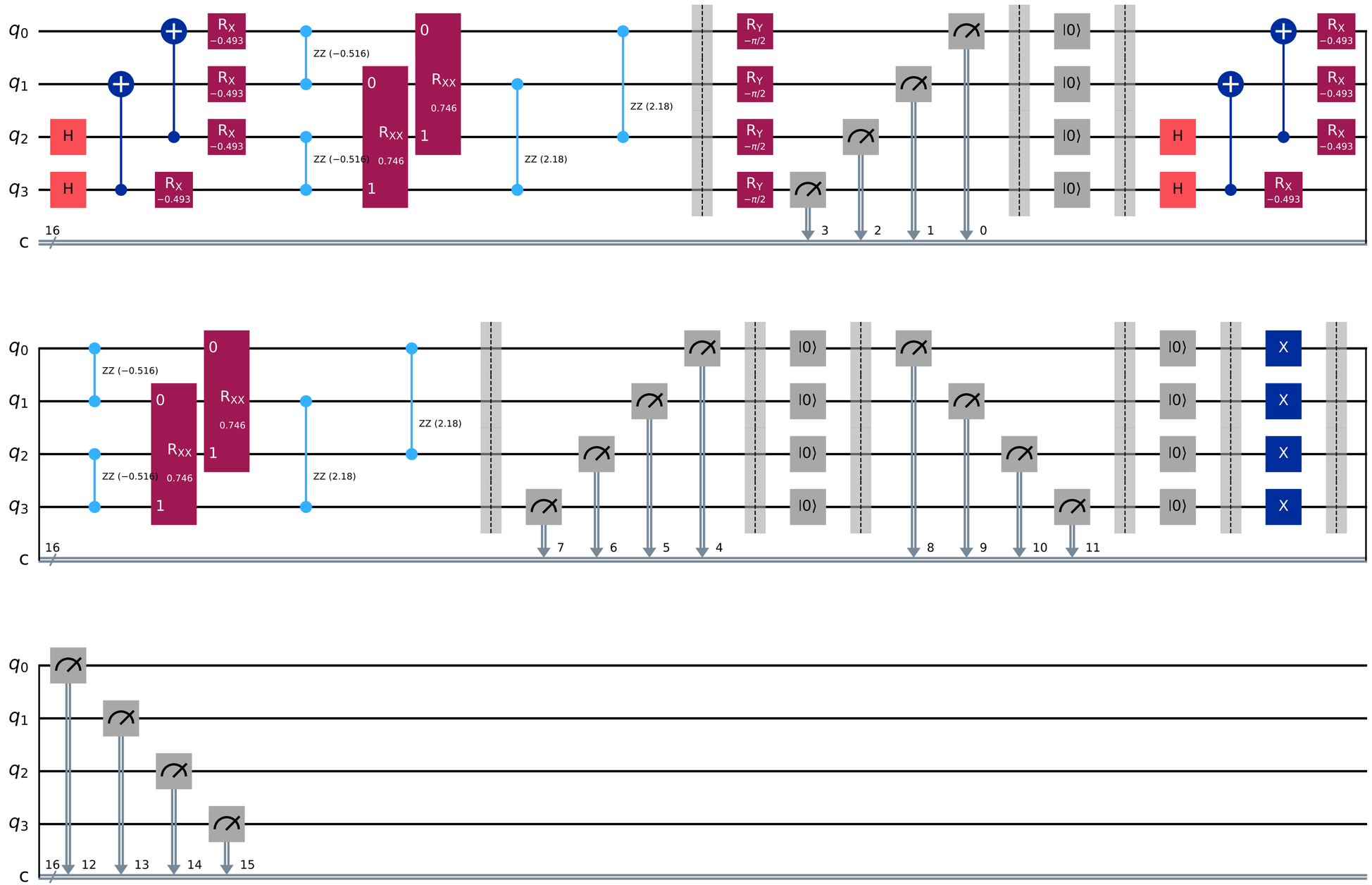
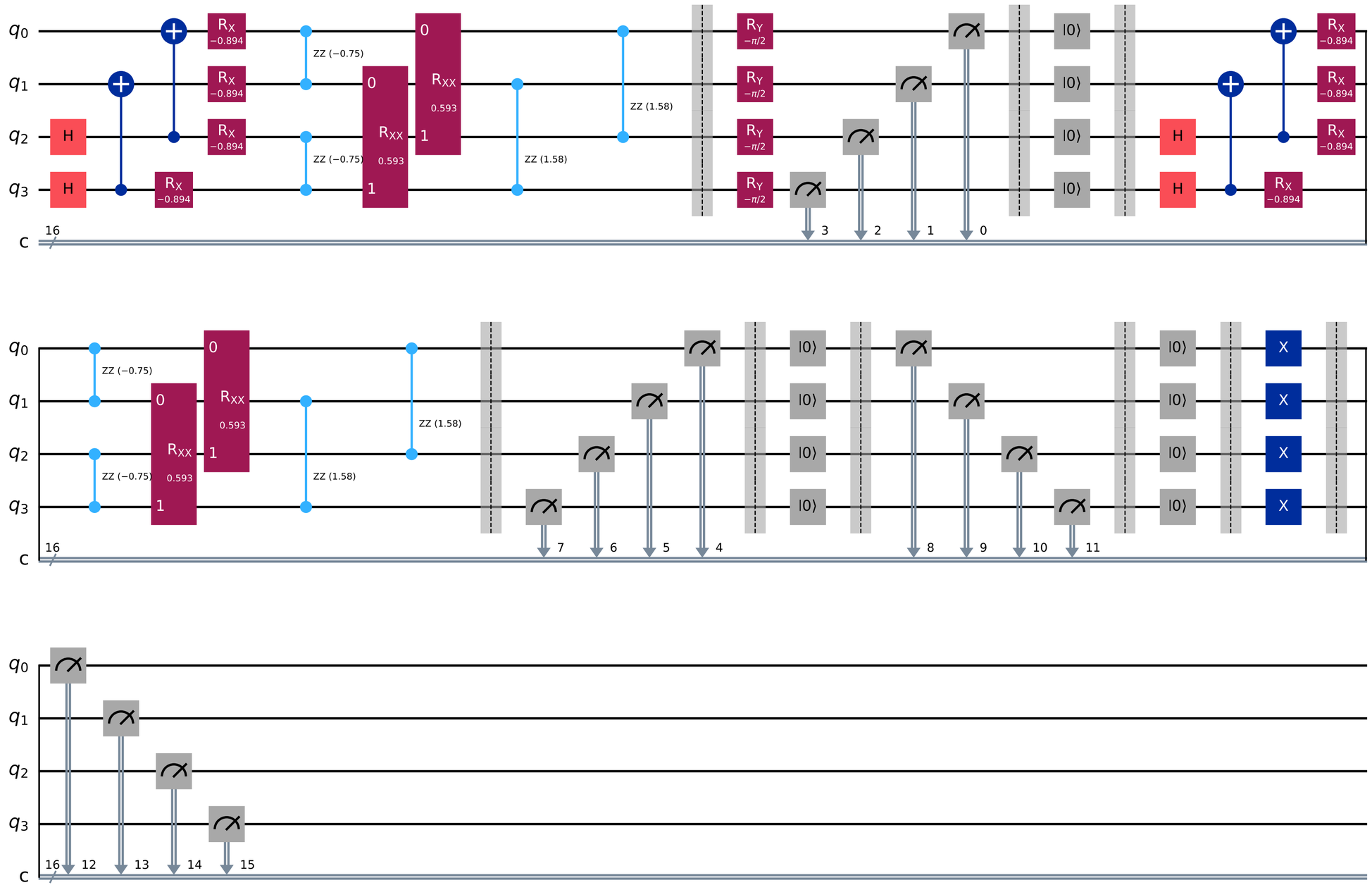

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.493
R_X −0.493
R_X −0.493
R_X −0.493
ZZ (−0.516)
ZZ (−0.516)
R_{XX} 0.746
ZZ (1.68)
ZZ (1.68)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.493
R_X −0.493
R_X −0.493
R_X −0.493
ZZ (−0.516)
ZZ (−0.516)
R_{XX} 0.746
ZZ (1.68)
ZZ (1.68)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

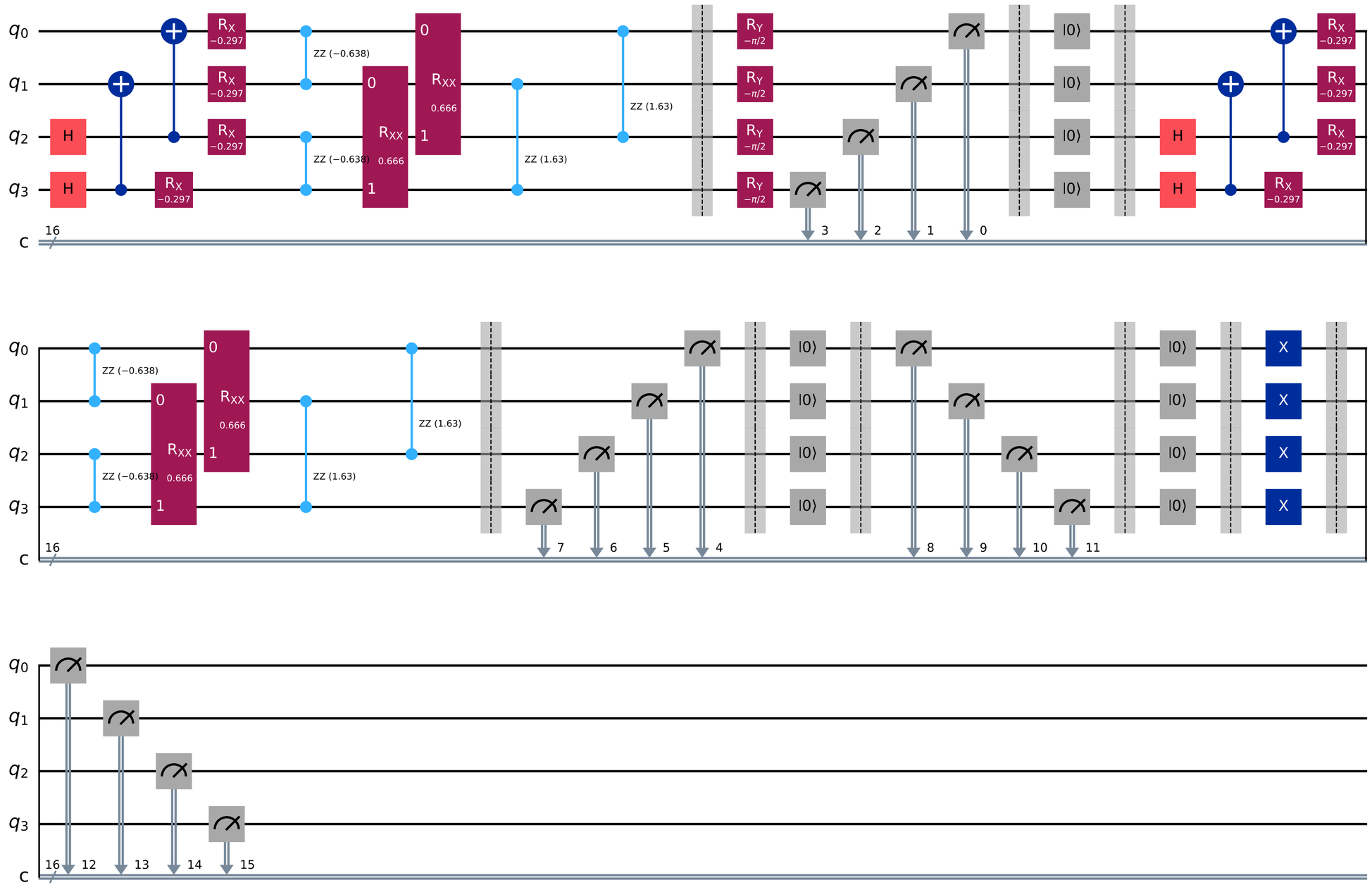

q0
q1
q2
q3
c
16
H
H
RX 0.00718
RX 0.00718
RX 0.00718
RX 0.00718
ZZ (−0.516)
ZZ (−0.516)
RXX 0.746
ZZ (1.68)
ZZ (1.68)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX 0.00718
RX 0.00718
RX 0.00718
RX 0.00718
ZZ (−0.516)
ZZ (−0.516)
RXX 0.746
ZZ (1.68)
ZZ (1.68)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15

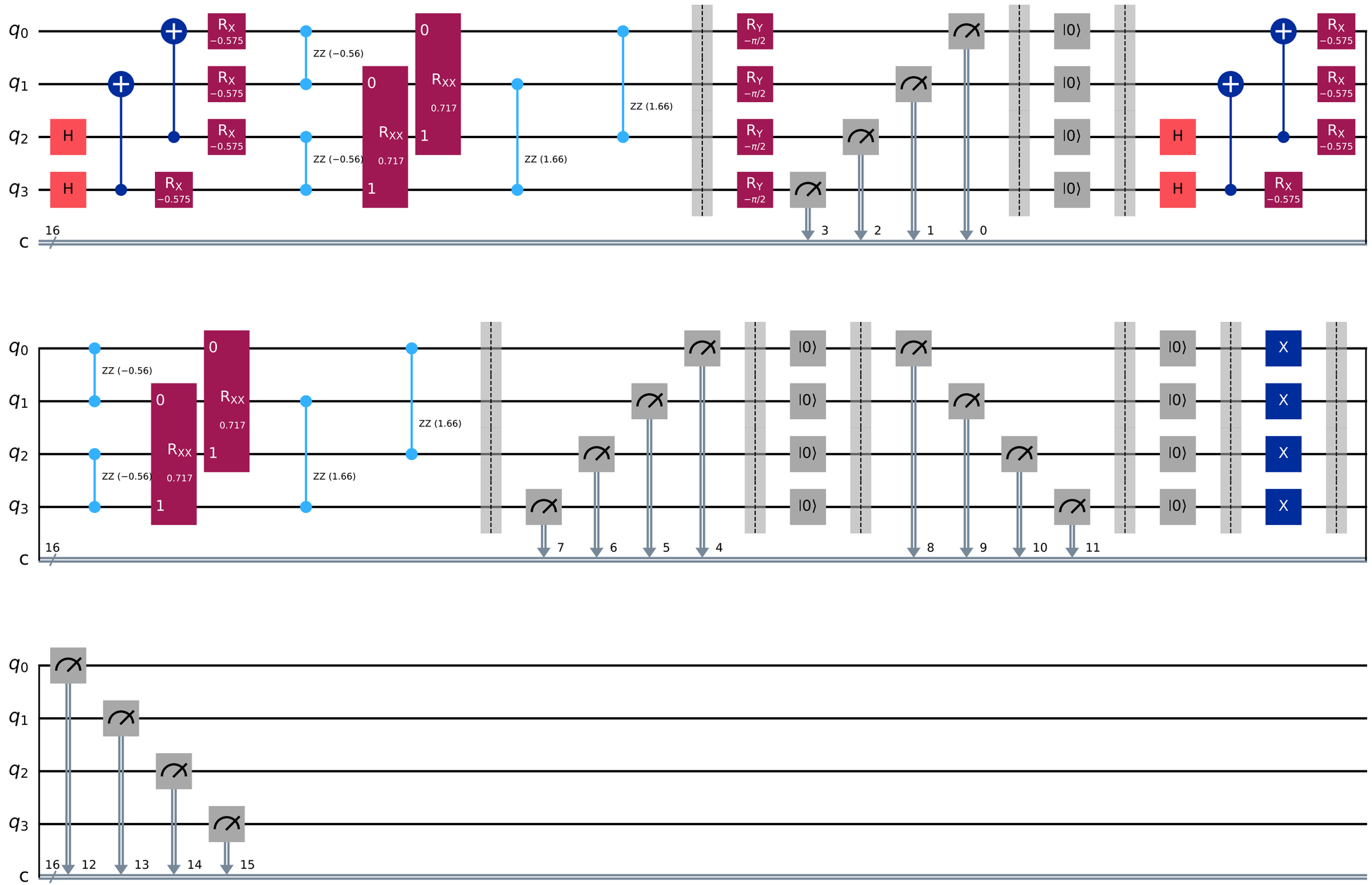
q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.493
R_X −0.493
R_X −0.493
R_X −0.493
ZZ (−0.0161)
ZZ (−0.0161)
0
R_{XX}
0.746
1
0
R_{XX}
0.746
1
ZZ (1.68)
ZZ (1.68)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.493
3
2
1
0
R_X −0.493
R_X −0.493
R_X −0.493
ZZ (−0.0161)
ZZ (−0.0161)
0
R_{XX}
0.746
1
ZZ (1.68)
ZZ (1.68)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

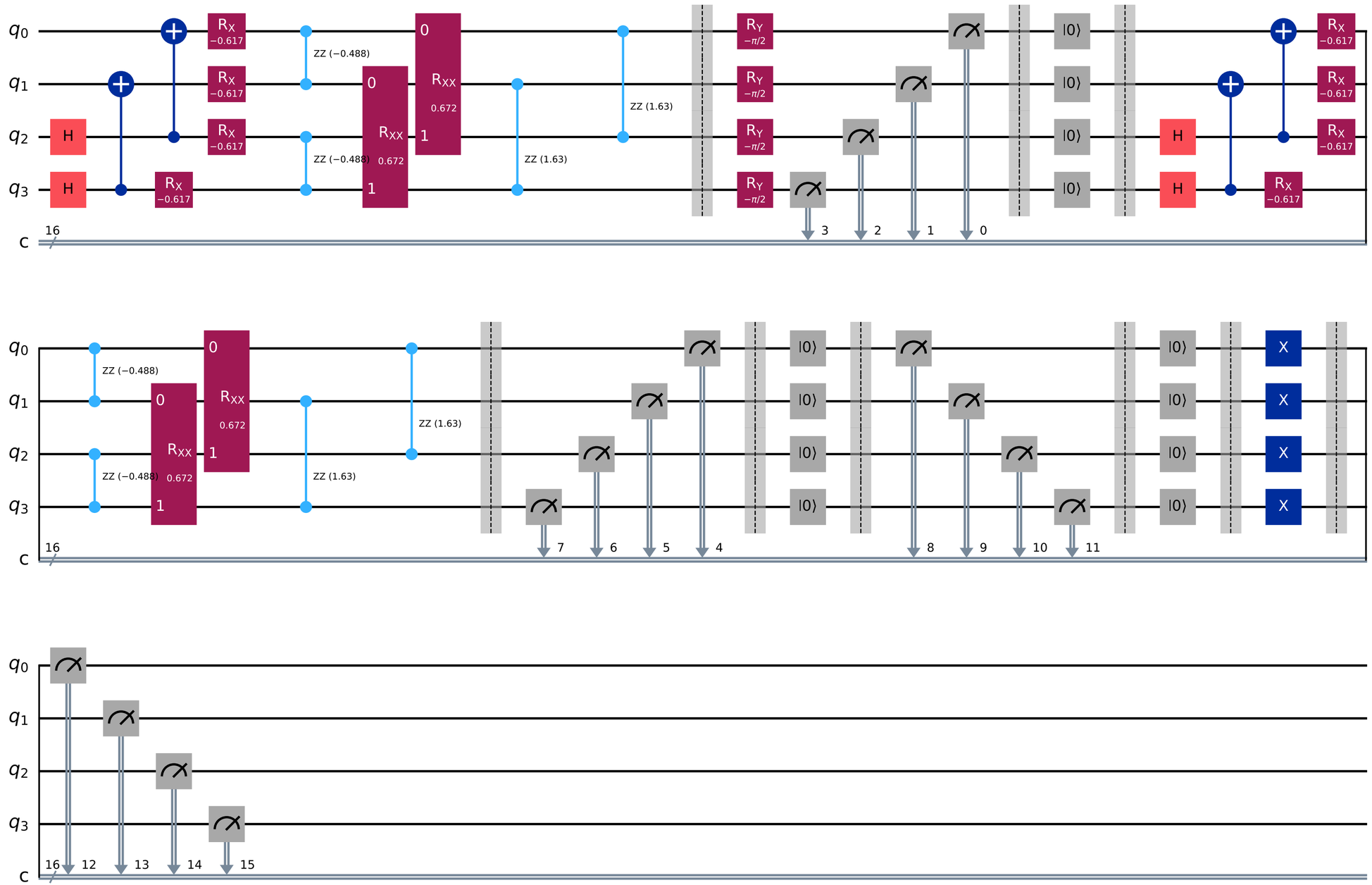

q0
q1
q2
q3
c
16
H
H
RX −0.493
RX −0.493
RX −0.493
RX −0.493
ZZ (−0.516)
ZZ (−0.516)
0
RXX 1.25
1
0
RXX 1.25
1
ZZ (1.68)
ZZ (1.68)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.493
RX −0.493
RX −0.493
ZZ (−0.516)
ZZ (−0.516)
0
RXX 1.25
1
0
RXX 1.25
1
ZZ (1.68)
ZZ (1.68)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

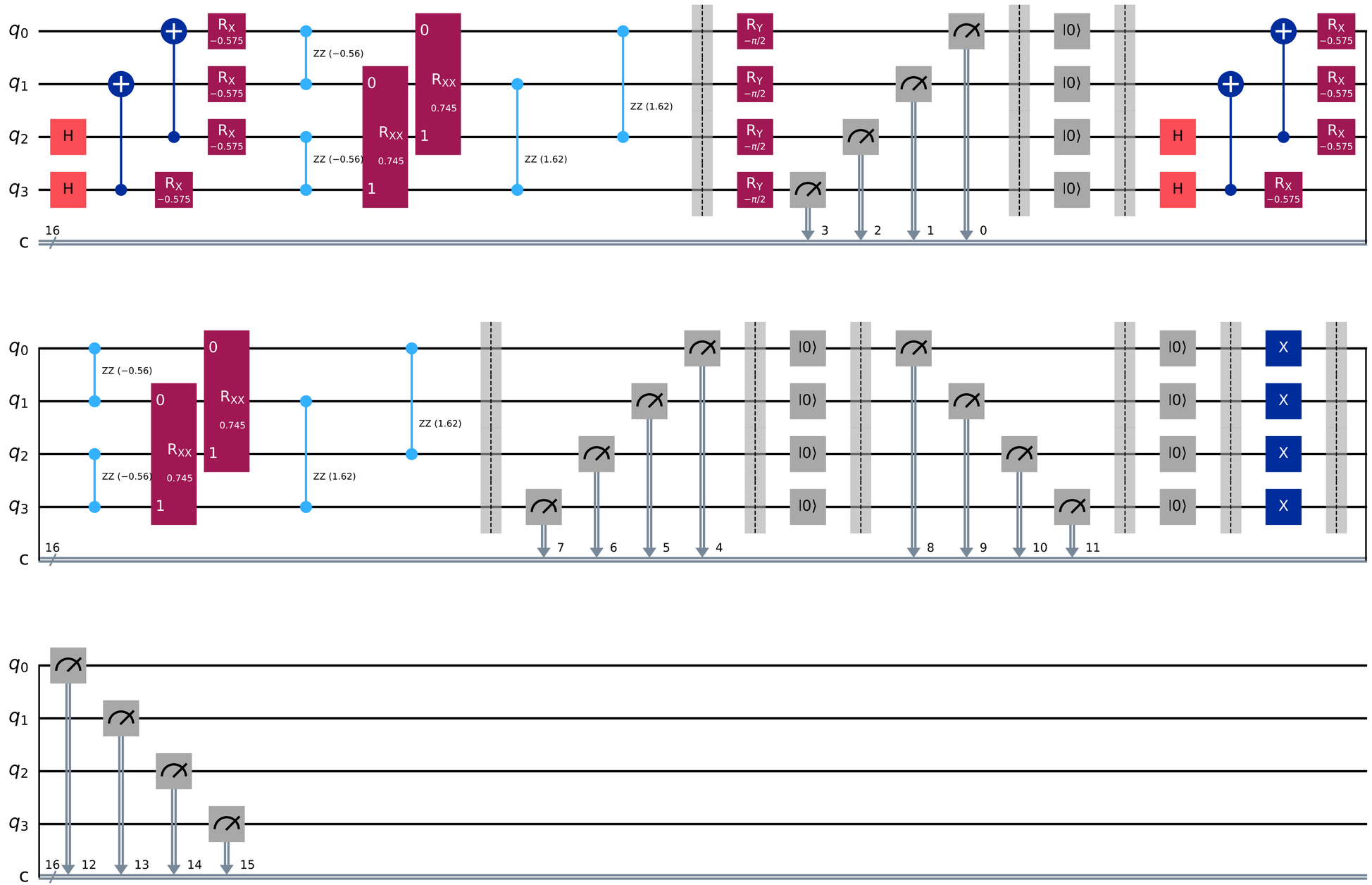

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.493
R_X −0.493
R_X −0.493
R_X −0.493
ZZ (−0.516)
ZZ (−0.516)
R_{XX} 0.746
ZZ (2.18)
ZZ (2.18)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.493
R_X −0.493
R_X −0.493
R_X −0.493
3
2
1
0
ZZ (−0.516)
ZZ (−0.516)
R_{XX} 0.746
ZZ (2.18)
ZZ (2.18)
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
|0⟩
X
X
X
X
7
6
5
4
8
9
10
11
12
13
14
15

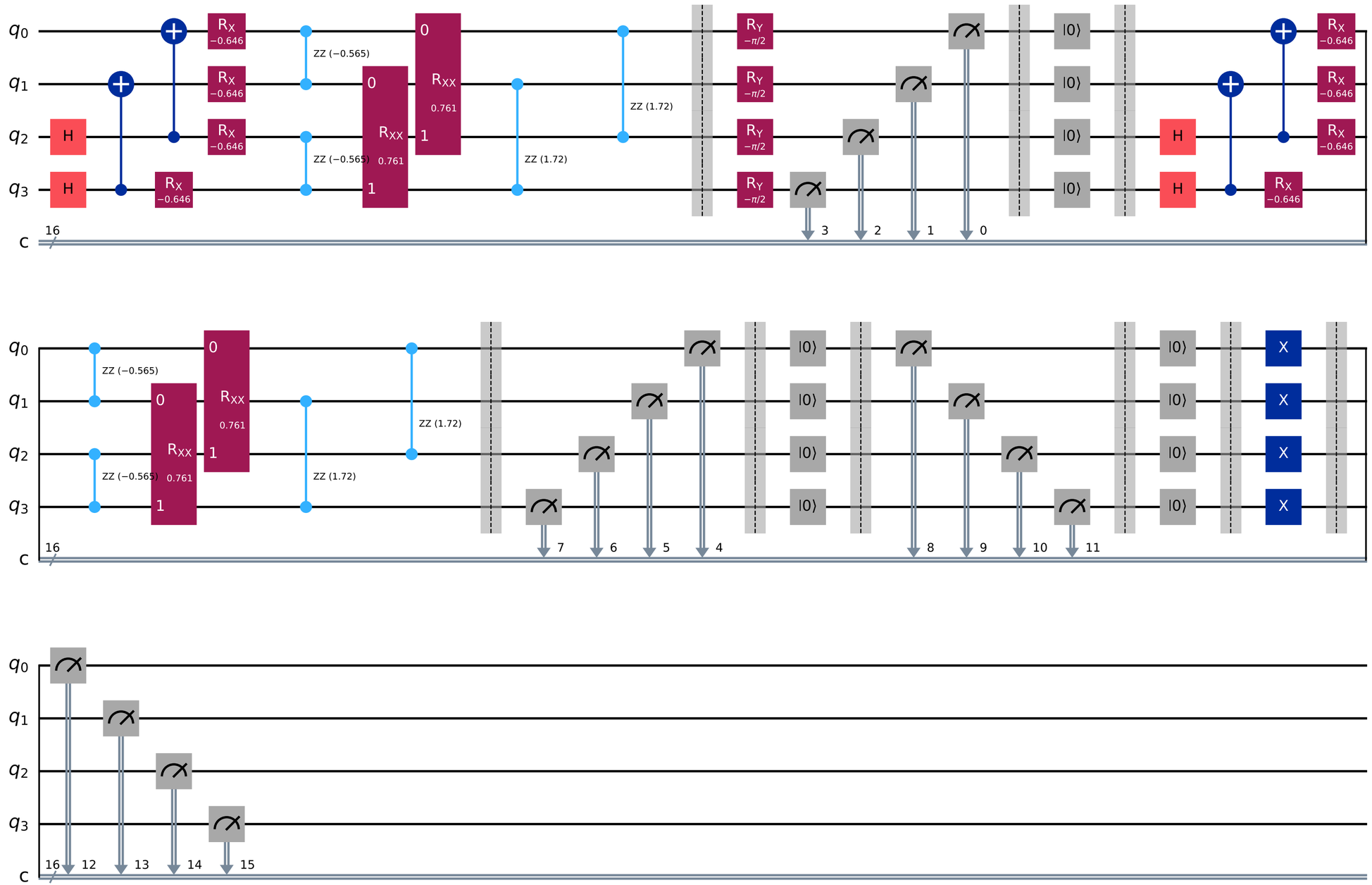

q0
q1
q2
q3
c
16
H
H
RX −0.894
RX −0.894
RX −0.894
RX −0.894
ZZ (−0.75)
ZZ (−0.75)
0
RXX
0.593
1
ZZ (1.58)
ZZ (1.58)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.894
RX −0.894
RX −0.894
RX −0.894
ZZ (−0.75)
ZZ (−0.75)
ZZ (1.58)
ZZ (1.58)
7
6
5
4
8
9
10
11
X
X
X
X
12
13
14
15

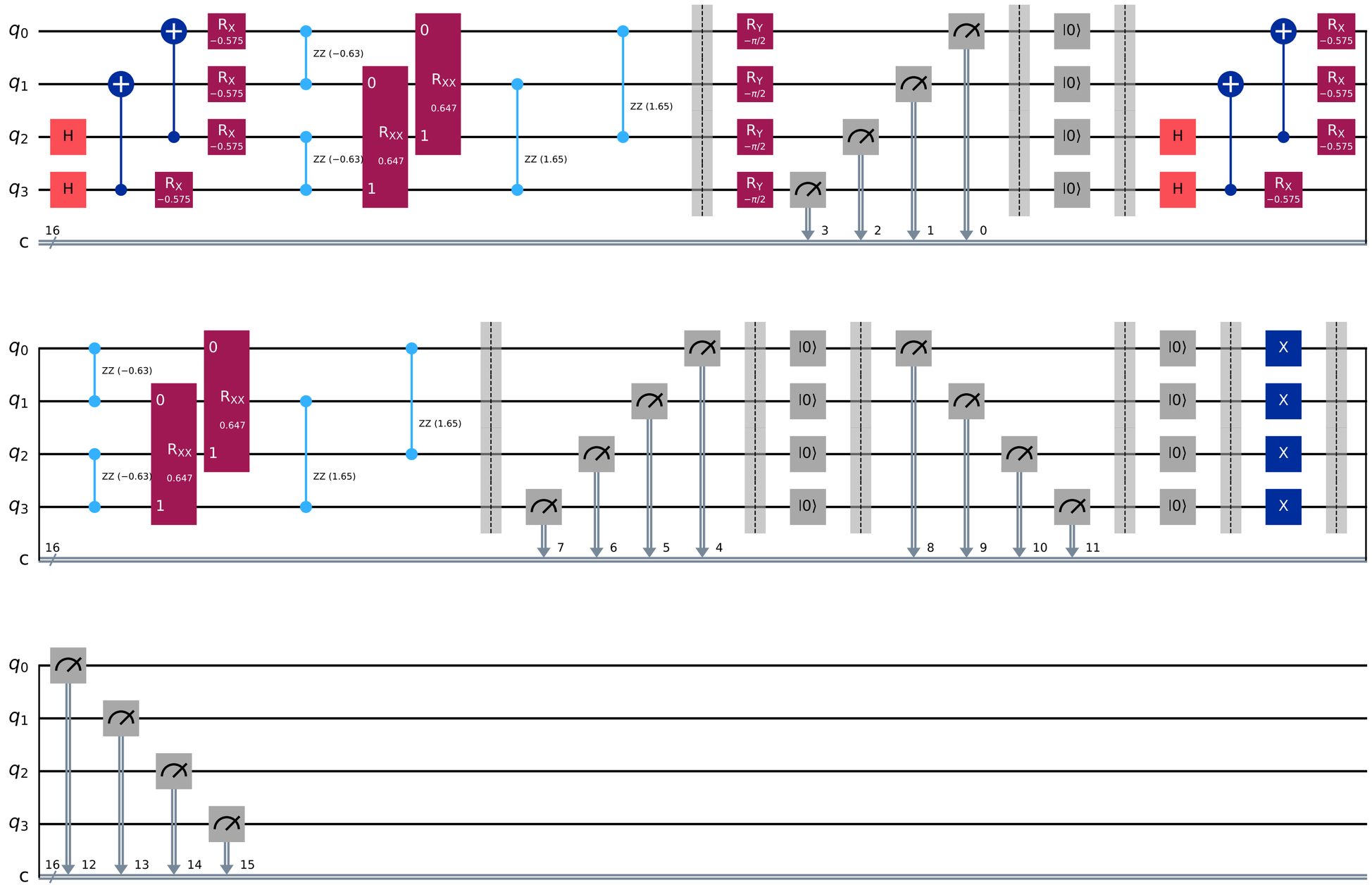

q0
q1
q2
q3
c
16
H
H
RX −0.297
RX −0.297
RX −0.297
RX −0.297
ZZ (−0.638)
ZZ (−0.638)
0
RXX
0.666
1
ZZ (1.63)
ZZ (1.63)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.297
RX −0.297
RX −0.297
RX −0.297
ZZ (−0.638)
ZZ (−0.638)
0
RXX
0.666
1
ZZ (1.63)
ZZ (1.63)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

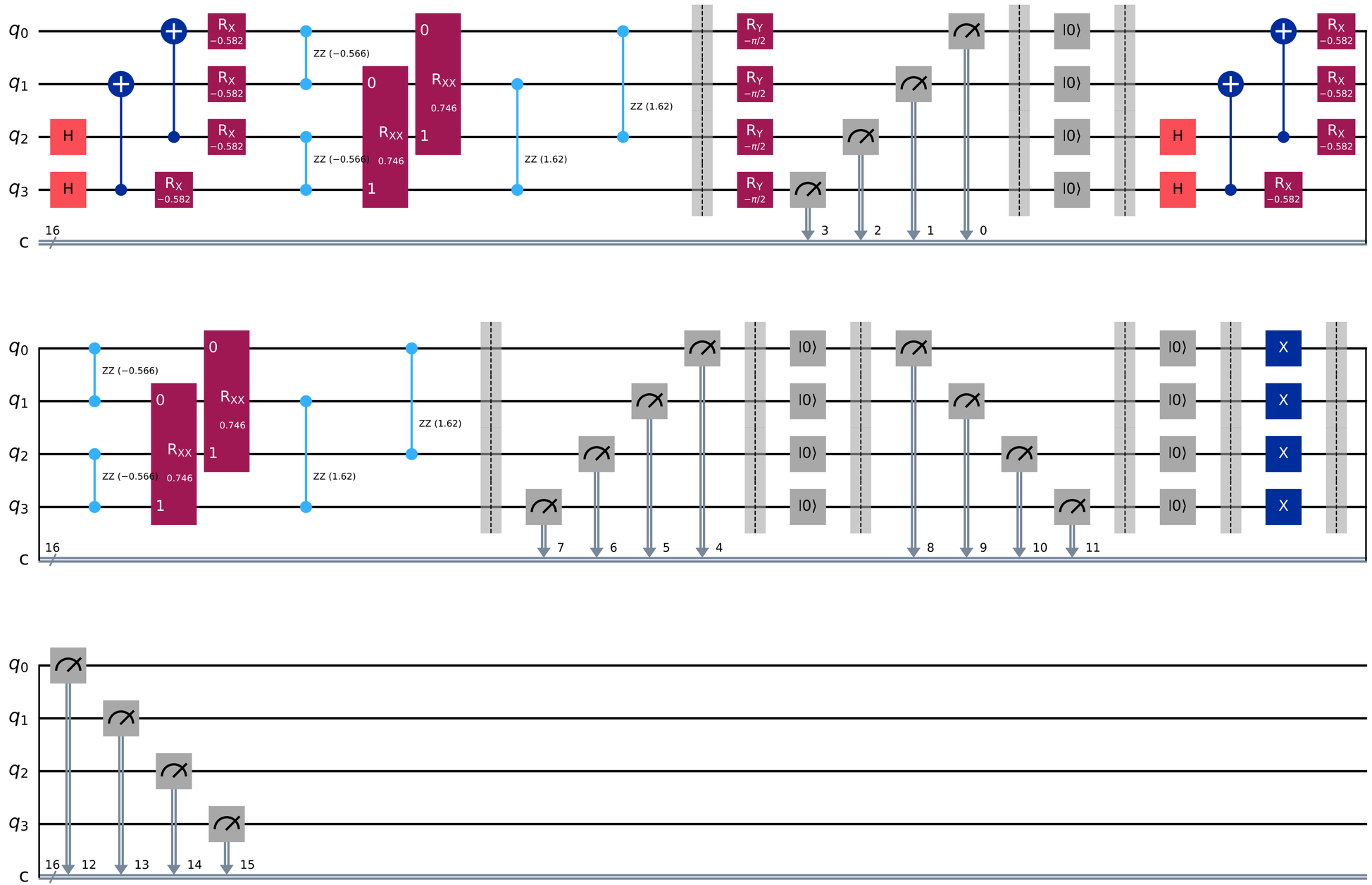

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.575
R_X −0.575
R_X −0.575
R_X −0.575
ZZ (−0.56)
ZZ (−0.56)
R_{XX} 0.717
ZZ (1.66)
ZZ (1.66)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.575
R_X −0.575
R_X −0.575
R_X −0.575
ZZ (−0.56)
ZZ (−0.56)
R_{XX} 0.717
ZZ (1.66)
ZZ (1.66)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

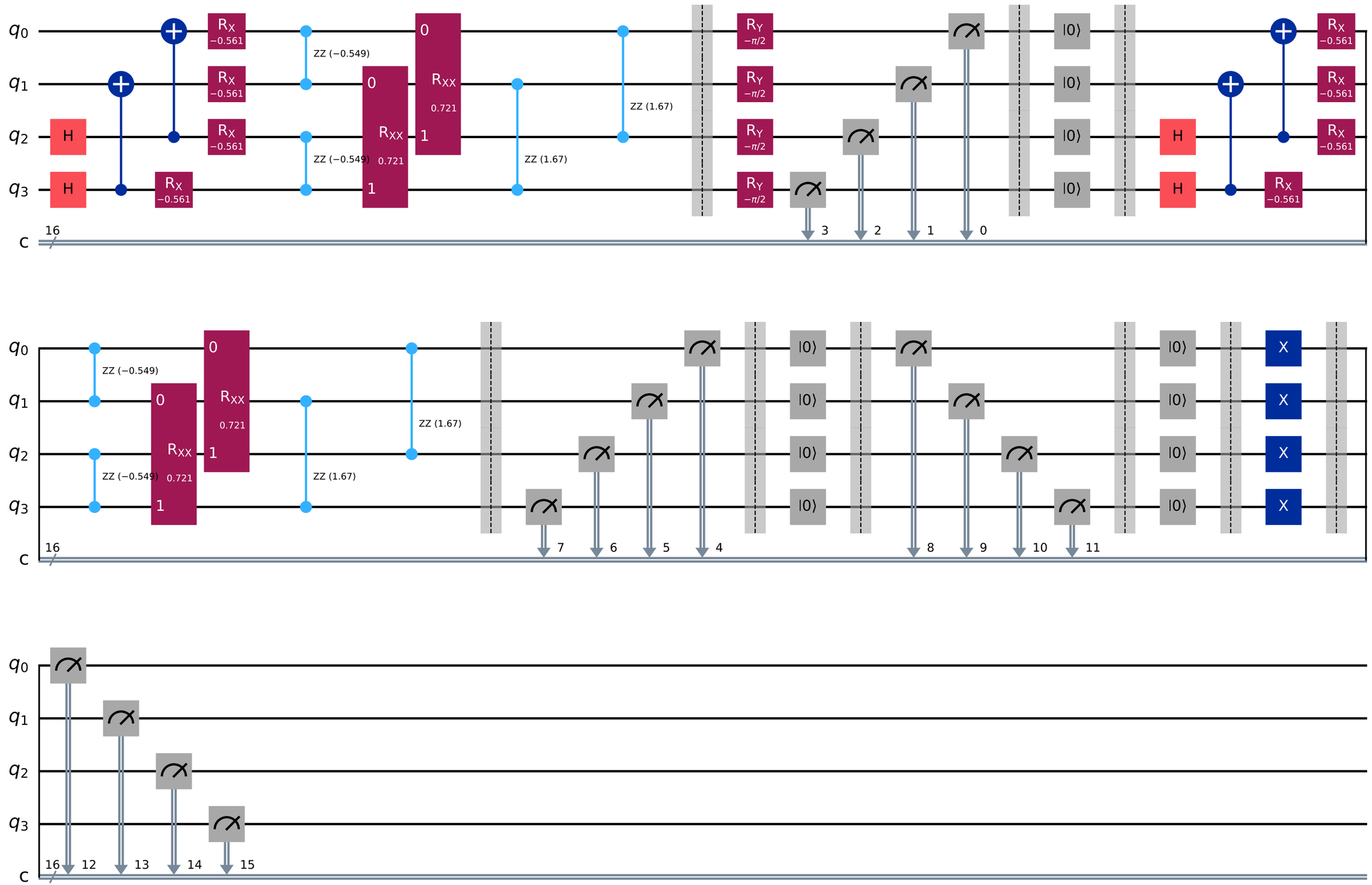

q0
q1
q2
q3
c
16
H
H
RX −0.617
RX −0.617
RX −0.617
RX −0.617
ZZ (−0.488)
ZZ (−0.488)
0
RXX 0.672
1
ZZ (1.63)
ZZ (1.63)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.617
RX −0.617
RX −0.617
RX −0.617
ZZ (−0.488)
ZZ (−0.488)
0
RXX 0.672
1
ZZ (1.63)
ZZ (1.63)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

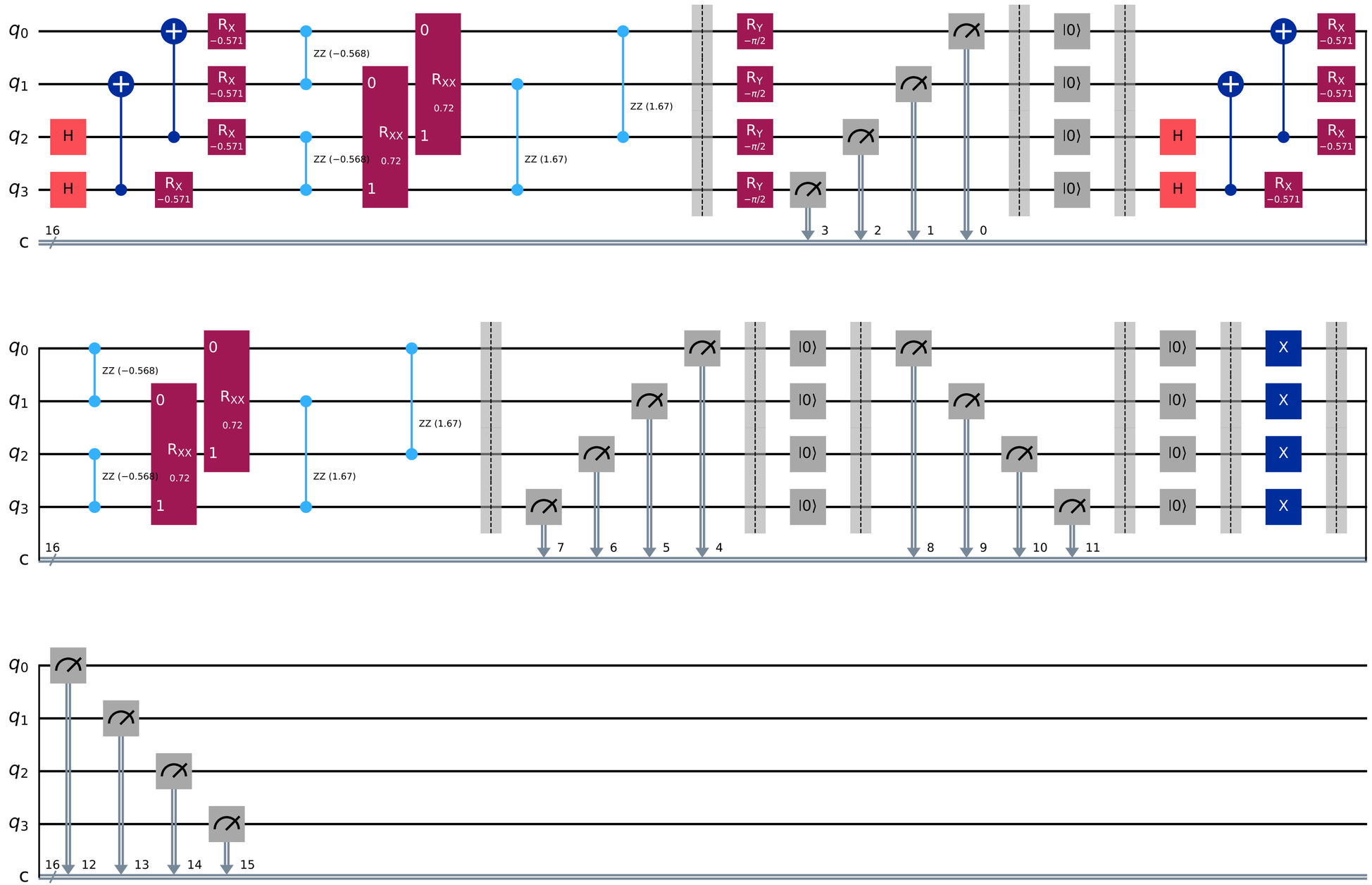

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.575
R_X −0.575
R_X −0.575
R_X −0.575
ZZ (−0.56)
ZZ (−0.56)
R_{XX} 0.745
ZZ (1.62)
ZZ (1.62)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.575
R_X −0.575
R_X −0.575
R_X −0.575
ZZ (−0.56)
ZZ (−0.56)
R_{XX} 0.745
ZZ (1.62)
ZZ (1.62)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

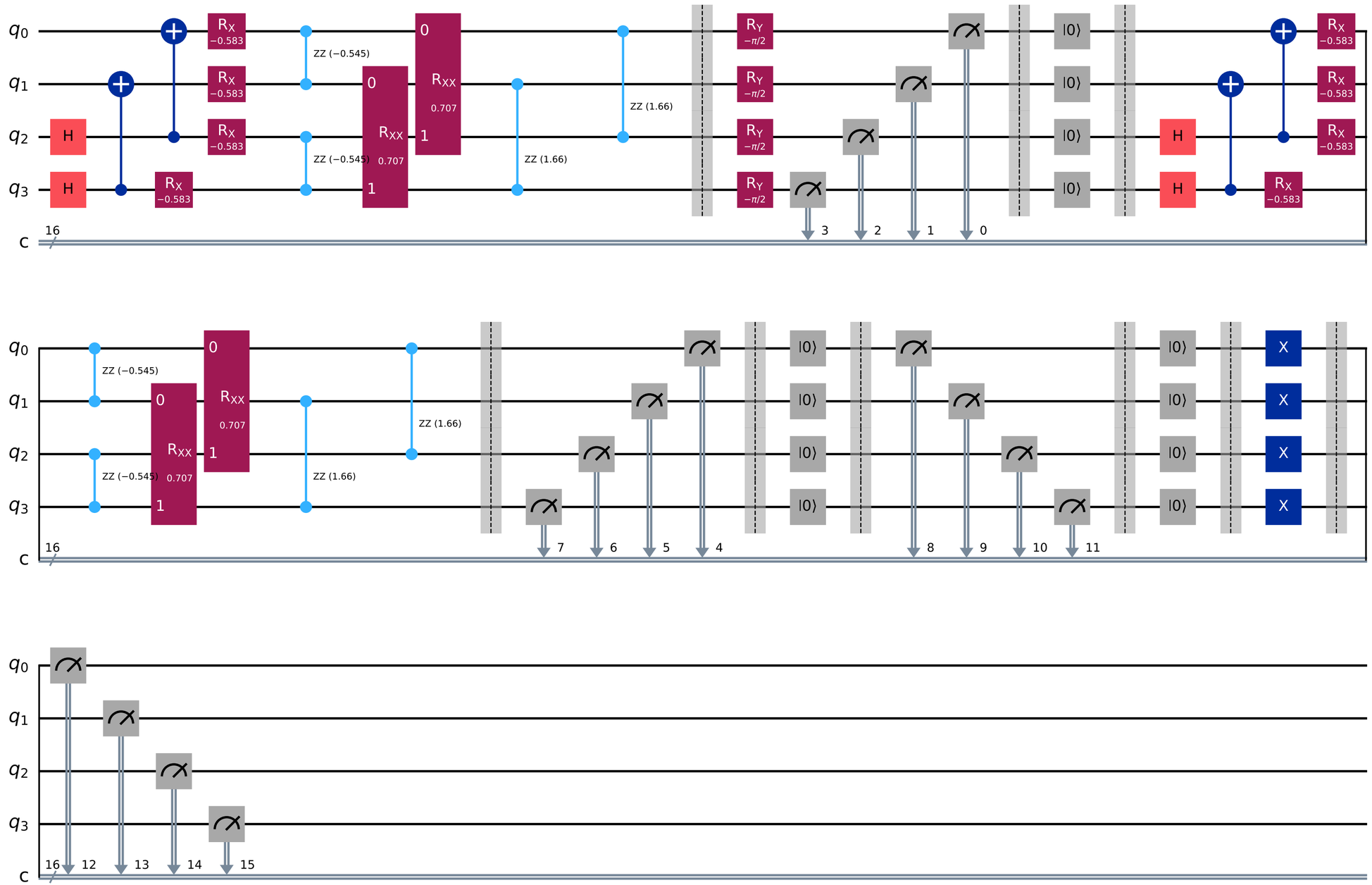

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.646
R_X −0.646
R_X −0.646
R_X −0.646
ZZ (−0.565)
ZZ (−0.565)
0
R_{XX} 0.761
1
ZZ (1.72)
ZZ (1.72)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.646
R_X −0.646
R_X −0.646
R_X −0.646
ZZ (−0.565)
ZZ (−0.565)
0
R_{XX} 0.761
1
ZZ (1.72)
ZZ (1.72)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

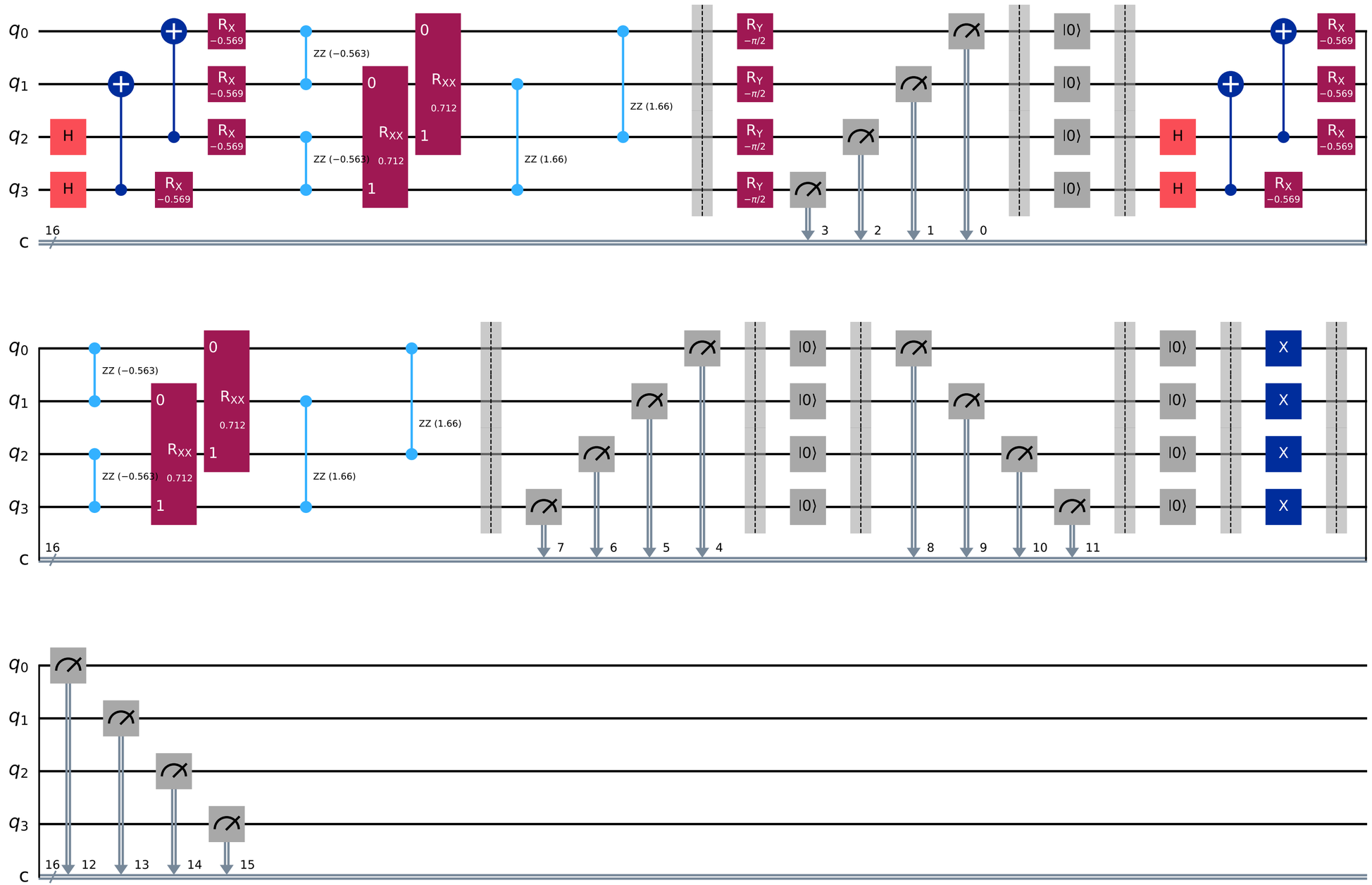

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.575
R_X −0.575
R_X −0.575
R_X −0.575
ZZ (−0.63)
ZZ (−0.63)
R_{XX} 0.647
ZZ (1.65)
ZZ (1.65)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.575
R_X −0.575
R_X −0.575
R_X −0.575
ZZ (−0.63)
ZZ (−0.63)
R_{XX} 0.647
ZZ (1.65)
ZZ (1.65)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

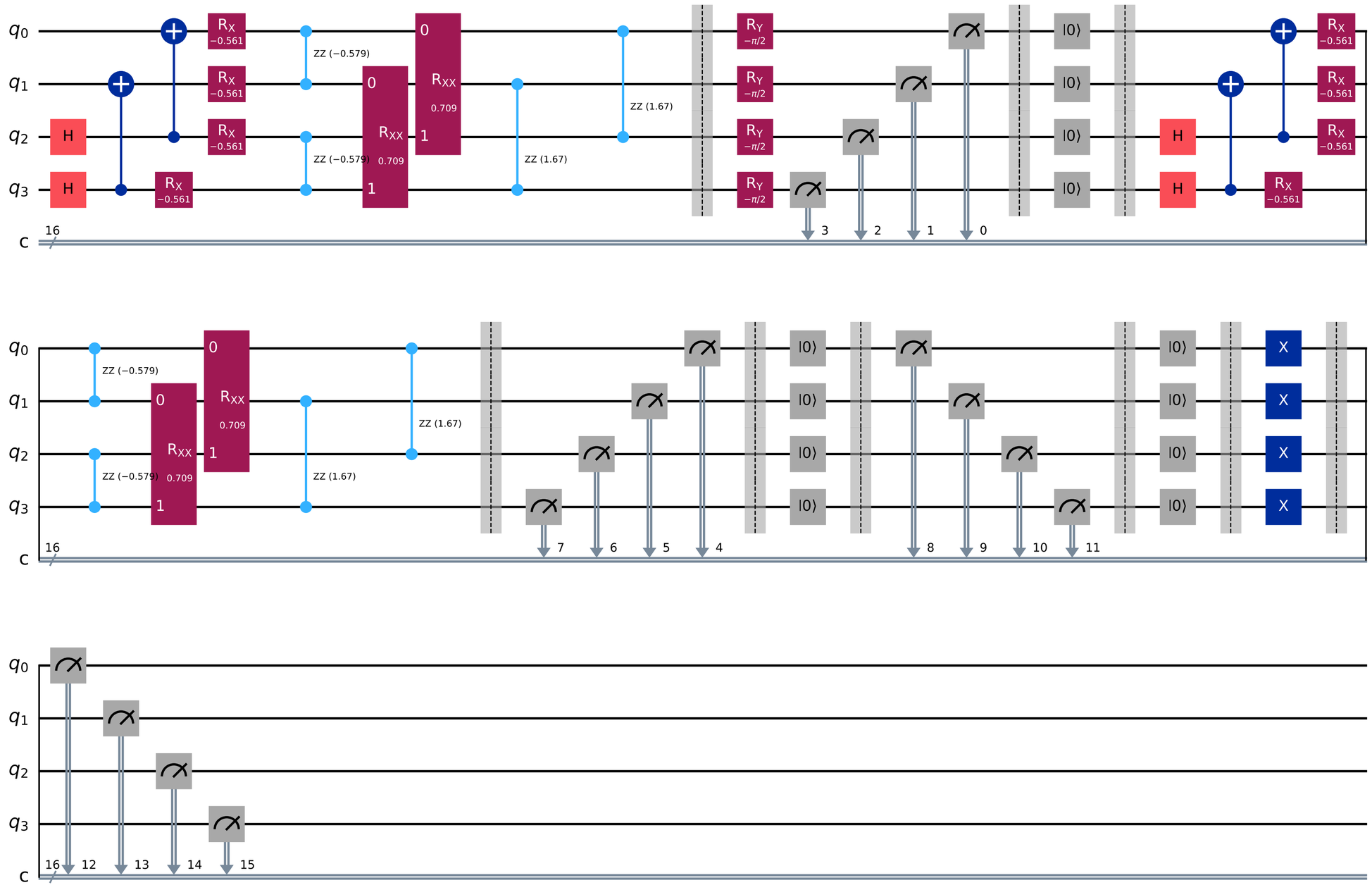

q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.582
R_X −0.582
R_X −0.582
R_X −0.582
ZZ (−0.566)
ZZ (−0.566)
R_{XX} 0.746
ZZ (1.62)
ZZ (1.62)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.582
R_X −0.582
R_X −0.582
R_X −0.582
ZZ (−0.566)
ZZ (−0.566)
R_{XX} 0.746
ZZ (1.62)
ZZ (1.62)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15

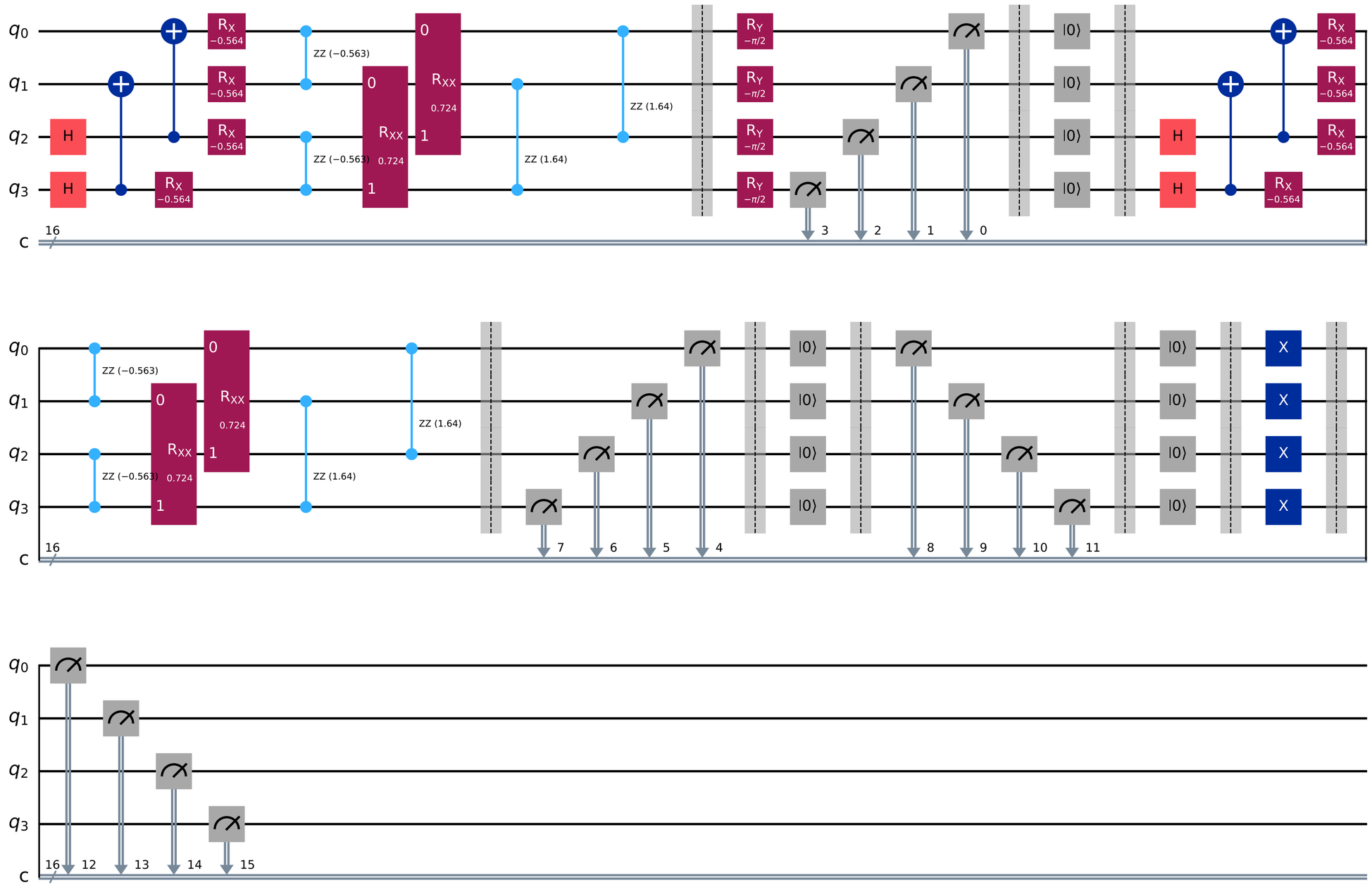

q0
q1
q2
q3
c
16
H
H
RX −0.561
RX −0.561
RX −0.561
RX −0.561
ZZ (−0.549)
ZZ (−0.549)
0
RXX 0.721
1
0
RXX 0.721
1
ZZ (1.67)
ZZ (1.67)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.561
RX −0.561
RX −0.561
ZZ (−0.549)
ZZ (−0.549)
0
RXX 0.721
1
0
RXX 0.721
1
ZZ (1.67)
ZZ (1.67)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15


q_0
q_1
q_2
q_3
c
16
H
H
R_X −0.571
R_X −0.571
R_X −0.571
R_X −0.571
ZZ (−0.568)
ZZ (−0.568)
R_{XX} 0.72
ZZ (1.67)
ZZ (1.67)
R_Y −π/2
R_Y −π/2
R_Y −π/2
R_Y −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
R_X −0.571
R_X −0.571
R_X −0.571
R_X −0.571
ZZ (−0.568)
ZZ (−0.568)
R_{XX} 0.72
ZZ (1.67)
ZZ (1.67)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.583
RX −0.583
RX −0.583
RX −0.583
ZZ (−0.545)
ZZ (−0.545)
0
RXX 0.707
1
0
RXX 0.707
1
ZZ (1.66)
ZZ (1.66)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.583
RX −0.583
RX −0.583
RX −0.583
ZZ (−0.545)
ZZ (−0.545)
0
RXX 0.707
1
0
RXX 0.707
1
ZZ (1.66)
ZZ (1.66)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
16
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.569
RX −0.569
RX −0.569
RX −0.569
ZZ (−0.563)
ZZ (−0.563)
0
RXX 0.712
1
ZZ (1.66)
ZZ (1.66)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.569
RX −0.569
RX −0.569
ZZ (−0.563)
ZZ (−0.563)
0
RXX 0.712
1
ZZ (1.66)
ZZ (1.66)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.561
RX −0.561
RX −0.561
RX −0.561
ZZ (−0.579)
ZZ (−0.579)
0
RXX 0.709
1
ZZ (1.67)
ZZ (1.67)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.561
RX −0.561
RX −0.561
RX −0.561
ZZ (−0.579)
ZZ (−0.579)
0
RXX 0.709
1
ZZ (1.67)
ZZ (1.67)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15


q0
q1
q2
q3
c
16
H
H
RX −0.564
RX −0.564
RX −0.564
RX −0.564
ZZ (−0.563)
ZZ (−0.563)
0
RXX
0.724
1
ZZ (1.64)
ZZ (1.64)
RY −π/2
RY −π/2
RY −π/2
RY −π/2
3
2
1
0
|0⟩
|0⟩
|0⟩
|0⟩
H
H
RX −0.564
RX −0.564
RX −0.564
RX −0.564
ZZ (−0.563)
ZZ (−0.563)
0
RXX
0.724
1
ZZ (1.64)
ZZ (1.64)
7
6
5
4
|0⟩
|0⟩
|0⟩
|0⟩
8
9
10
11
|0⟩
|0⟩
X
X
X
X
12
13
14
15